DFU-WAGNER Net: Complete Multi-Task End-to-End Architecture

Input Image x

RGB $224 \times 224 \times 3$

Conv Stream

EfficientNetV2-S

FusedMBConv / Local Features
Output: $1,280$-D

CNN Projection

Linear(1280, 1280)
+ BN + GELU + Drop(0.3)

1,280-D

ViT Stream

Swin-Transformer Tiny

Shifted Windows / Global Topology
Output: 768-D

ViT Projection

Linear(768, 1280)
+ BN + GELU + Drop(0.3)

1,280-D

Fused Features

$z_{\text{fused}} \in \mathbb{R}^{2,560}$

2,560-D

Hard-Pair LMCoT

Tangent Spherical ($\mathcal{L}_{\text{lmcot}}$)

TaskDisentangler Module

$z_{\text{path_private}}$

$\mathbb{R}^{512}$ (L_2 Normalized)

$\mathcal{L}_{\text{orth}} \perp$

z_{shared}

$\mathbb{R}^{768}$ (L_2 Normalized)

$\mathcal{L}_{\text{orth}} \perp$

$z_{\text{depth_private}}$

$\mathbb{R}^{512}$ (L_2 Normalized)

$z_s \oplus z_p$

Pathology Head

CB-Focal Loss
4 Classes (p_{path})

$z_s \oplus z_d$

Wagner Depth Head

CORAL Ordinal Loss
Grades 0--4 (p_{depth})

5×4 **Joint Matrix**

Exploratory Bivariate Risk

Uncertainty-Gated MMD

Domain Alignment ($\mathcal{L}_{\text{mmd}}$)